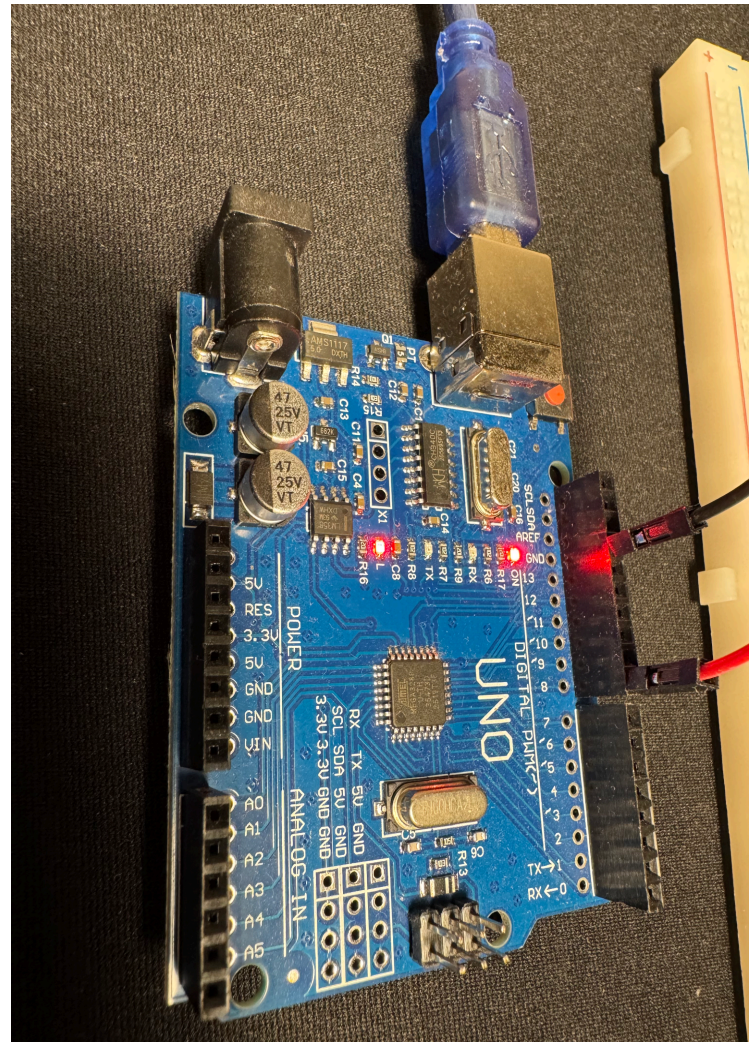
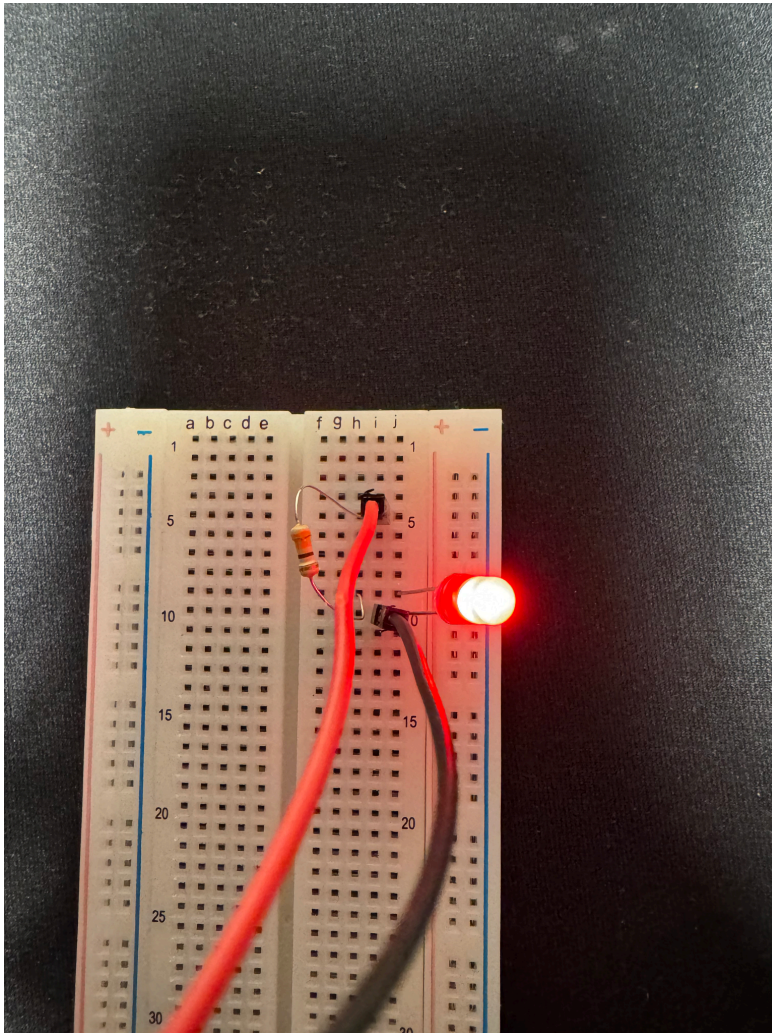
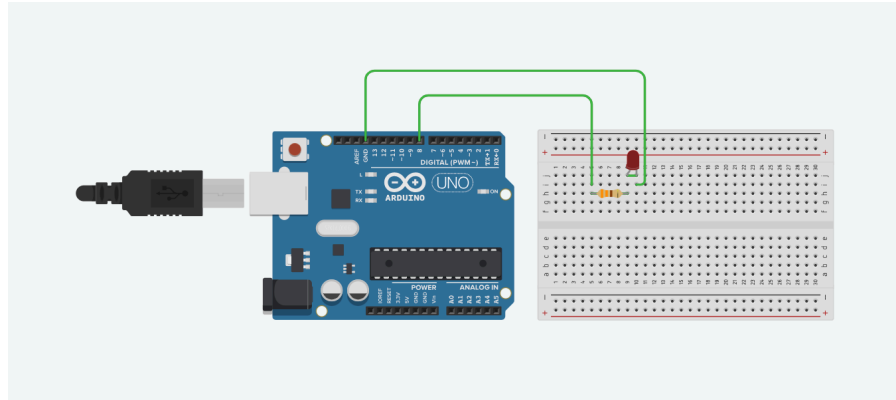


Blink



Circuit Components

- LED (light-emitting diode)

LEDs typically operate at around 15mA (milliampere). Applying 5V directly may damage the LED, so it must be protected with a resistor.

Ohm's Law: $V = I \times R$

$V \rightarrow$ Voltage (Volt (V))

$I \rightarrow$ Current (Ampere (A))

$R \rightarrow$ Resistance (Ohm (Ω))

- Resistor

LEDs cannot tolerate excessive current. Therefore, a 330 Ω resistor is used to protect the LED.

Basically we can state like this: $R = V / I \rightarrow 5 / 0.015 = 333.33\Omega$

However the resistor calculation must include the LED's forward voltage:

$$R = (V_s - V_f) / I$$

Where:

- $V_s \rightarrow$ Arduino output voltage

- $V_f \rightarrow$ LED forward voltage

- $I \rightarrow$ Desired LED current

- $R \rightarrow$ Required resistance

$$V_s = 5 \text{ V}, \quad V_f = 2 \text{ V}, \quad I = 15 \text{ mA} = 0.015 \text{ A}$$

$$R = (5-2) / 0.015 = 200 \Omega$$

A standard 220 Ω resistor would be suitable. However, this circuit uses a 330 Ω resistor to provide additional current limiting.

The approximate current with a 330 Ω resistor is:

$$I = (5-2)/330 = 0.0091 \text{ A} = 9.1 \text{ mA}$$

Therefore, the 330 Ω resistor safely limits the current and protects the LED.

The calculated value is approximate because the actual forward voltage varies depending on the LED color and model. In this circuit the color of the LED is red that is why we got it as 2 here.

- Breadboard

A breadboard is a reusable plastic block filled with small holes used to build and test electronic circuits without soldering.

- Arduino

A printed circuit board that houses the microcontroller chip alongside extra supportive parts like power regulators, USB communication interfaces and easy-to-use pin headers.

- Jumper Wires

Jumper wires are short electrical wires with a pin or connector at each end, used to join two points in a circuit without soldering.

Circuit Connection

1. Connect the LED's short leg (cathode, -) directly to one of the Arduino's GND pins using a jumper wire.
2. Connect the LED's long leg (anode, +) to Arduino pin 8 through a 330Ω resistor.

What Happens?

- When pin 8 is HIGH, it outputs about 5V and the LED turns on.
- When pin 8 is LOW, it outputs 0V and the LED turns off.
- `delay(500)` pauses the program for 500 milliseconds which is half a second.

Pin 8 → 330Ω resistor → LED's long leg (anode) → LED → short leg (cathode) → GND

Code

```
#define red_led 8

void setup() {
  // put your setup code here, to run once:
  pinMode(red_led, OUTPUT); // We just say to Arduino that pin 8 is an
  output port.
}
```

```
void loop() {  
    // put your main code here, to run repeatedly:  
    digitalWrite(red_led, HIGH); // We send a HIGH signal (5V) to pin 8. So,  
the LED turns on.  
    delay(500); // Wait for 500ms. In other words, Let 500ms pass while the  
LED is on.  
    digitalWrite(red_led, LOW); // We send a LOW signal (0V) to pin 8. So,  
the LED turns off.  
    delay(500); // Wait for 500ms. In other words, Let 500ms pass while the  
LED is off.  
}
```